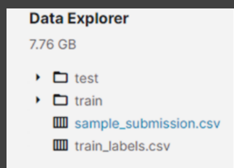
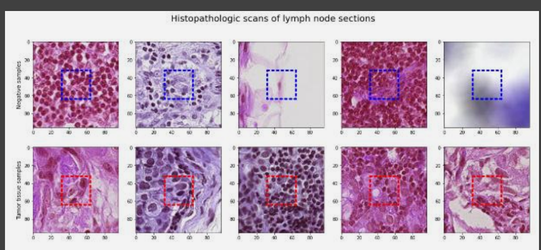
Histopathologic Cancer Detection



Problem Statement

- Early Cancer Detection can ⇧ Treatment Success Rate
- Metastatic Cancer is difficult for doctors to identify
- Various imaging techniques for detection are used
- Manual Tumor Detection Approach

Our team will create a model that will help doctors to identify small cell cancers using digital pathology scans.

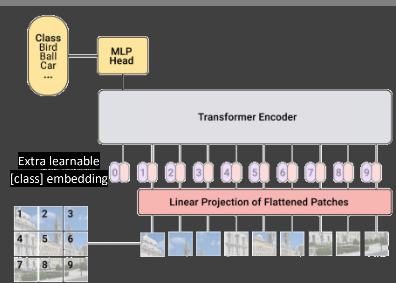


Project History

- 93% accuracy by using a Sequential Deep CNN.
- 57% accuracy by using ResNet CNN

Data

- Image Files have an image id name.
- We used the train_labels.csv file and train images to test and train our model.
- A positive label indicates that the whole image contains at least one pixel of tumor tissue.



Vision Transformer (ViT) Model

- Data Augmentation: Ability to create more data
- Improves upon CNNs (Strong Model Performance)
- Features
  - Input 16K Images
  - No Pre-trained ViT Model
  - Used RMSprop Optimizer
  - Patch Extraction for Pixel-level Analysis

	precision	recall	f1-score	support
0	0.79	0.90	0.84	1604
1	0.89	0.76	0.82	1596
accuracy			0.83	3200
macro avg	0.84	0.83	0.83	3200
weighted avg	0.84	0.83	0.83	3200

Results

- Accuracy: 83%
- F-Score: 83%
- Precision: 84%
- Recall/Sensitivity: 83%

Opportunities

- ⇧ Number of images & epochs, ⇧ Accuracy
- Pre-trained ViT Model Exploration & Use